Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/04/2016
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Project Title:	Synergistic Effects of Bcl2, Cigarette Smoking and Space Radiation on Carcinogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30322-1013	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	12/12/2011	**End Date:**	12/11/2015
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/11/2015 per NSSC information (Ed., 12/16/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX12AC30G		
Performance Goal No.:			
Performance Goal Text:			
	Space radiation, comprised of energetic protons and heavy nuclei, has been shown to produce distinct biological damage compared with radiation on Earth, leading to large uncertainties in the projection of cancer. Space radiation can induce various types of DNA lesions, including DNA single-strand breaks (SSBs), double-strand DNA breaks (DSBs), and others. Specific DSBs may lead to chromosomal breaks or translocations that contribute to genomic instability and/or carcinogenesis. It is known that terrestrial radiation is largely characterized by low linear energy transfer (LET) radiation (i.e., x-, beta-, or gamma-rays). By contrast, space radiation is characterized by high-LET radiation (i.e., high energy protons and high charge (Z) and energy (E) (HZE) nuclei). Bcl2 not only functions as a potent antiapoptotic molecule but also as a cellular proto-oncogene that can promote tumor development. Our recent studies indicate that Bcl2 suppresses the repair of terrestrial radiation, i.e., low-LET gamma-ray radiation-induced DSBs, which lead to an		

Task Description:	accumulation of DNA damage with increased chromosomal breaks in living cells. Additionally, we found that Bcl2 also inhibits the repair of abasic sites of DNA lesions and impedes DNA mismatch repair. However, it is currently unclear whether Bcl2 promotes space radiation-initiated carcinogenesis via suppressing the repair of space radiation-induced DNA damage. The repair of DSBs is known to occur mainly through the non-homologous end-joining (NHEJ) and homologous recombination (HR) pathways. Bcl2 has been reported to suppress the RAD51-dependent recombination pathway. We recently found that Bcl2 potently inhibits the NHEJ pathway via downregulation of Ku DNA binding and DNA-PKcs activities in human lung cancer cells. Additionally, Bcl2 also inhibits the repair of abasic sites of DNA lesions and impedes DNA mismatch repair. These properties of Bcl2 may contribute to increased genetic instability and development of various malignancies, including lung cancer. Based on our previous findings, we hypothesize that Bcl2 may impede the repair of space radiation-induced DNA damage in normal cells, which could enhance susceptibility to carcinogenesis. Major radiation-induced solid cancer sites include breast, thyroid, colon, and lung. Lung cancer makes up about one-third of the cancers attributable to radiation observed in atomic-bomb survivors and nuclear-reactor workers. Because lung cancer has a strong etiological association with cigarette smoking, astronauts with active or passive exposure to cigarette smoking may be at higher risk of lung cancer. Since nicotine is a major component in cigarette smoke that can activate Bcl2 via phosphorylation, we hypothesize that space radiation may synergize with nicotine to facilitate lung cancer development, at least in part, in a Bcl2-dependent manner. To critically test these hypotheses, two specific aims have been identified: (1) To determine whether and how Bcl2 suppresses the repair of high-LET radiation-induced double-strand DNA breaks (DSBs) leading to enhanced genetic instability; (2) To determine whether high-LET radiation and nicotine act synergistically in the development of lung cancer. Studies will test whether nicotine synergizes with high-LET radiation to facilitate the malignant transformation of normal lung bronchial epithelial cells, and whether selective expression of Bcl2 in lung tissues enhances susceptibility to lung carcinogenesis following treatment with high-LET radiation alone or in combination with nicotine. Results from our proposed studies are expected to contribute significantly to the development of novel strategies for prevention and/or treatment of space radiation-and/or-cigarette smoking-induced lung cancer or other Bcl2-expressing malignancies by specifically disrupting the oncogenic activity of Bcl2.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	My research will benefit developing new strategies to prevent and/or treat space radiation and cigarette smoking-induced lung cancer by targeting oncoprotein Bcl2.
Task Progress:	Closeout Report Introduction The major objectives of this project are to evaluate whether and how Bcl2 negatively regulates the repair of high-LET radiation-induced DNA damage in normal lung epithelial cells, to test whether Bcl2 enhances high-LET radiation-induced genetic instability, to demonstrate whether selective expression of Bcl2 in lung tissues increases susceptibility to carcinogenesis following high-LET radiation exposure, and to determine whether high-LET radiation and nicotine synergistically facilitate the malignant transformation of normal lung bronchial epithelial cells in vitro and the formation of lung cancer in vivo. Specific Aim 1: To determine whether and how Bcl2 suppresses the repair of high-LET radiation-induced double-strand DNA breaks (DSBs) leading to enhanced genetic instability. Specific Aim 2: To determine whether high-LET radiation and nicotine act synergistically in the development of lung cancer. Studies will test whether nicotine synergizes with high-LET radiation to facilitate the malignant transformation of normal lung bronchial epithelial cells, and whether selective expression of Bcl2 in lung tissues enhances susceptibility to lung carcinogenesis following treatment with high-LET radiation alone or in combination with nicotine. Summary of Research Results obtained during grant period (12-12-2011 to 12-11-2015): High-LET ionizing radiation (IR) derived from high charge (Z) and energy (E) (HZE) particles has been reported to inhibit the Ku-dependent nonhomologous end-joining (NHEJ) pathway by small DNA fragments but does not inhibit the homologous recombination (HR) DNA repair pathway. Thus, HR pathway may play a major role in repairing DSBs induced by high-LET IR. The Mre11 complex (Mre11/Rad50/NBS1)-mediated resection of DSB ends is a required step in preparing for DSB repair via the HR pathway. We found that expression of Bcl2 inhibits the repair of the HZE particles (i.e., 56Fe and 37Si)-induced DSBs by inhibiting Mre11 complex activity. Exposure of cells to 56Fe or 37Si promotes Bcl2 to interact with Mre11 via the BH1 and BH4 domains. Purified Bcl2 protein directly suppresses Mre11 complex-mediated DNA resection in vitro. Expression of Bcl2 reduces the ability of Mre11 to bind DNA following exposure of cells to HZE particles. Our findings suggest that, after cellular exposure to HZE particles, Bcl2 may inhibit the Mre11complex-mediated DNA resection leading to suppression of the HR-mediated DSB repair in surviving cells, which may contribute to tumor development. These findings have been published in “Nucleic Acids Research” in 2015. For animal work, we generated lung-targeting WT Bcl2 transgenic mice under CC10 promoter. Mice were exposed to high-LET radiation derived from 56Fe, 28Si, or 16O at Brookhaven National Laboratory (BNL) in the absence or presence nicotine as proposed. Results show that high-LET radiation and nicotine synergistically facilitate the formation of lung cancer in lung-targeting WT Bcl2 transgenic mice. The manuscript is under preparation and we plan to submit to “Cancer Discovery” Journal for publication. Publications resulting from my last NASA grant: 1. Wang H, Wang X, Chen G, Zhang X, Tang X, Park D, Cucinotta FA, Yu DS, Deng X, Dynan WS, Doetsch PW, Wang Y. (2014) Distinct roles of Ape1 protein, an enzyme involved in DNA repair, in high or low linear energy transfer ionizing radiation-induced cell killing. J Biol Chem. 289, 30635-44. PMCID: PMC4215242 . 2. Xie M., Park D., You S., Li R., Owonikoko T.K., Wang Y., Paul W. Doetsch P.W., and Deng X. (2015) Bcl2 inhibits recruitment of Mre11 complex to DNA double-strand breaks in response to high-linear energy transfer radiation. Nucleic Acids Res. 43, 960-72. PMCID: PMC4333404 .

	3. Xie M., Park D., Li R., and Deng X. Bcl2, nicotine and space radiation synergistically promotes lung carcinogenesis. Cancer Discovery (In preparation).
Bibliography Type:	Description: (Last Updated: 11/12/2020)
Articles in Other Journals or Periodicals	Wang H, Wang X, Chen G, Zhang X, Tang X, Park D, Cucinotta FA, Yu DS, Deng X, Dynan WS, Doetsch PW, Wang Y. "Distinct roles of Ape1 protein, an enzyme involved in DNA repair, in high or low linear energy transfer ionizing radiation-induced cell killing." J Biol Chem. 2014 Oct 31;289(44):30635-44. Epub 2014 Sep 10. http://dx.doi.org/10.1074/jbc.M114.604959 ; PubMed PMID: 25210033; PubMed Central PMCID: PMC4215242. ED. NOTE: Article re-categorized as "Other" since was WITHDRAWN May 2020, see PMID: 32358082; PMCID: PMC7196659 , Oct-2014
Articles in Peer-reviewed Journals	Xie M, Park D, Sica GL, Deng X. "Bcl2-induced DNA replication stress promotes lung carcinogenesis in response to space radiation." Carcinogenesis. 2020 Mar 11;bgaa021. Published: 11 March 2020. https://doi.org/10.1093/carcin/bgaa021 ; PMID: 32157295 , Mar-2020